2014 3rd International Conference on User Science and Engineering (i-USEr)

A Study of Smartphone Usage and Barriers Among the Elderly

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Abstract—**A wide range of smartphone applications aimed at encouraging healthy diets and active lifestyles have been developed in recent years. However, the use of these smartphone applications among the elderly population that should benefit the most from them remains low. This paper reports on the barriers to smartphone usage among the elderly. In-depth interviews were conducted with the elderly to gather information about their demographics, experiences and the perceived barriers to using smartphones. Twenty-one participants were interviewed using a semi-structured interview method. Results show that the elderly have difficulties in using smartphones due to multiple reasons such as financial limitations, vision impairments, and lack of interest and knowledge in using technological devices and their advanced functionalities.**

Keywords – smartphone usage; barriers; elderly

I. INTRODUCTION

A vast amount of research has demonstrated that the use of assistive technologies and information communication technologies (ICTs) have the potential to enable older adults to improve their quality of life and reduce healthcare costs [1], [2]. These include the development of various smartphone applications aimed at encouraging a healthy lifestyle. Researchers also regard these applications as a great tool to support autonomous living while maintaining quality of life among the elderly as they experience declines in perceptual, motor, and cognitive abilities due to natural aging [3], [4].

Although there were similar studies conducted in [5] and [6], it's important to note that, during that time, smartphone were not popular as it is nowadays. This is shown by the fact that, even these earlier studies found that elderly experienced some difficulties with its user interface design, such as buttons and font size, they still choose to use conventional type of phone. Reference [7] also demonstrate similar attempt to understand smartphone usage among the elderly, but the study however mainly concentrated on investigating facilitating factors leading to smartphone adoption rather than inhibiting factors.

This study hence reports on the barriers of smartphone usage among the elderly. Despite of the fact that there is a wider selection of mobile phone to be chosen, we were enthralled to investigate why most elderly choose to use simple feature phone, which offered minimal functionalities instead of advanced functionalities offered by smartphone.

However, it is important to note that this study was mainly aim to understand elderly perceptions and perceived barriers of using the technology, and not focusing on investigating usability issues associated with particular smartphone application.

II. RELATED WORK

There have been only a few studies that investigated mobile phone usage specifically among the elderly. Reference [5] found that the main reason for using mobile phones among the elderly were for emergency, safety and security reasons. Short Message Service (SMS) has become the most used function, while video calls are the least used function. Reference [6] replicated this study in Malaysia and found similar findings i.e. preparing for emergency situations and having casual conversations with family and friends were the most common reasons for having a mobile phone among the elderly participants.

However, in terms of phone design, both studies indicated that apart from the small screen size, the elderly participants had difficulties with the mechanical buttons on their mobile phones, mentioning that those buttons were too small and were made of rubber. They preferred metallic buttons, which 'clicked when pressed' [5], [6]. These findings clearly indicate that these earlier studies focused on the usage of feature phones instead of smartphones as smartphones involve touchscreen interaction where pressing on metallic buttons is not possible.

Meanwhile, a comparative study by [3] on differences in the ways younger and older adults learn to use smartphones revealed that learning approaches and preferences vary across ages. Older adults prefer to learn through printed manual instructions, as compared to their younger counterparts who prefer a trial-and-error method. This indicates that despite the intuitive user interface offered by most smartphones that encourages self-exploration and easy recovery, older adults seem to lack confidence in this experimentation. This could perhaps be due to past negative experiences such as frustration over particular smartphone functionality that did not work the way they wanted, or their unintentional alteration of the phone configurations leading to malfunction. As elderly people commonly experience mild vision impairments, we are concerned that learning how to use smartphones by having to read instruction manuals which tend to use a tiny font size and

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unfamiliar technical jargon [8] might be one of the barriers to smartphone acceptance among elderly users.

According to a recent report by The Nielsen Company [9], which specializes in observing and measuring consumer markets, there are three types of mobile phones, which are feature phones, multimedia phones and smartphones as shown in Fig. 1.

Feature phones generally come with a much lower price tag, smaller screen size, low-resolution camera, keypad mechanical buttons with basic functionality such as making calls, sending and receiving text, taking pictures, and video messaging. Meanwhile, multimedia phones are mobile phones that can handle music, pictures and videos. They are often designed around one or a few major features such as a high-resolution camera or high quality sound. Smartphones, on the other hand, are the most recent and advanced mobile phones, but they are very pricey. They run on a dedicated mobile operating system such as Google's Android, Apple's iOS and Windows Phone from Microsoft. Basic features include touchscreen interface, WiFi connectivity, a high-resolution camera, integrated GPS, and a range of downloadable applications.

However, despite numerous potential benefits offered for the well being of the elderly, the chances of these smartphone-based applications reaching wider groups of the elderly population seem to be slim as a recent report by the Nielsen Company mentioned that younger users are more likely to own smartphones and older users are more likely to own feature phones [9].

This trend is further supported by similar findings by the Pew Research Centre, which found that only 18 percent of the elderly population in the United States owned a mobile phone compared to 81 percent of their younger counterparts [10]. These statistics may lead to the quick assumption that this phenomenon is due to the fact that most of the elderly aged 60 years and above are pensioners, have lower incomes and are hence unable to afford expensive smartphones. However, a study on smartphone penetration based on age and income revealed that younger adults, regardless of income level, are very likely to be smartphone owners even at the lowest income brackets [9]. Smith [10] suggested that this potentially indicates that young people consider a smartphone as a necessity rather than a luxury. The elderly, in contrast, regard smartphone ownership as non-essential and more of an "elite" phenomenon.

Fig. 1. Types of mobile phone [9]

Similar findings were found in a study conducted among Malaysian elderly [7], which indicated that compared to younger adults in 40s who shift to smartphone technology due to self-demand or personal preferences, older adults of 60s however only feel the urges to use smartphone due to family encouragement [7].

A recent empirical study conducted by Culen and Bratteteig [11] on the use of touch interfaces among the elderly revealed that none of the elderly participants had experience with touch interfaces including that of the smartphone. Most of the elderly used feature phones, which involved the use of large buttons and straightforward functions [11].

III. METHODS

To understand the barriers in using the smartphone among the elderly population, a qualitative approach through semi-structured interviews was chosen, as we wanted to have an in-depth understanding of how mobile phones were being used in the elderly community.

A. Participants

Convenience sampling method was chosen, as there is no list of smartphone users available that make population sampling possible. Our study focused on older adults aged 60 and above. Apart from age, based on previous studies [3], [5], other inclusion criteria were that they had to have had experience with mobile phone usage for at least one year. This experience factor was important, as the objective of the study is not to identify usability problems in the mobile phone user interface but to seek opinion among the elderly who find having a mobile phone a necessity but refuse to use the more advanced smartphone.

Other exclusion criteria include severe cognitive, hearing, or vision impairments that make smartphone usage not possible. Cognitive health was assessed through the MMSE test, while hearing and vision were assessed through self-reported data.

Twenty-one elderly Malaysian of Malay, Chinese and Indian ethnicity participated in this study. All participants were recruited from the Senior Citizen Activity Centre (SCAC), a centre that provides facilities for Malaysians aged above 60; it conducts activities such as aerobics, cooking classes and social gatherings. Several SCACs have been established throughout Peninsular Malaysia, but for the purpose of this study only two SCACs located in Cheras Baru, Kuala Lumpur and Bukit Baru, Melaka were selected. These two locations were chosen, as both are located in urban areas that were assessable to the researcher and reportedly had greater numbers of the elderly with mobile phone experience.

We disseminated information about this study at the SCACs a week earlier, and those willing to participate signed up and were selected for participation based on their ability to meet our predefined set of inclusion and exclusion criteria for which they were assessed informally through self-reported data and were approved by their respective SCAC advisors. Participation in this project was voluntary and required a written consent before participation. All data were collected during a period of 1 month, from September to October 2013.

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B. Procedures

As the elderly are a vulnerable group of participants, and may experience some difficulties in mobility, all interview sessions were hence conducted at their respective senior centres. They were given an explanation on the objectives of the study, which are to identify issues, problems and challenges related to the use of smartphones among elderly Malaysians. Each participant was informed that the interview session would consist of three parts: demographic assessment (part A), current mobile phone usage pattern (part B), barriers to using smartphones (part C).

After the participants had consented to continue with the experiment, they were asked to complete the demographic assessment questionnaire, which was read out to them. After that, the interview session began. The session was audio-recorded for later analysis. Based on pilot studies, we had estimated that each interview session would require approximately 20 minutes to be completed.

C. Data Analysis

The first part elicited demographic information and comprised several questions about the attribute of each participant such as age, education background, employment history and prior experience with technology. The data were then analysed using descriptive statistics.

Meanwhile, the subsequent interview sessions on their usage of the mobile phone were audio-recorded, and were later transcribed and manually analysed using a thematic analysis technique.

IV. Results and Discussions

The following sections summarize demographic information of participants involved in this study.

A. Demography

Out of twenty-one participants, twelve of the elderly used feature phones, three used multimedia phones and six used smartphones. As expected, all six smartphone users were from the younger age group: three from the 60-64 age group and another three from the 65-69 age group. None of the participants aged 70 years and above reported using smartphones.

To understand factors influencing their chosen type of phone, we investigated various factors, mainly education, employment history and technological experiences. In terms of education background, the majority of feature phone users (seven out of 12 elderly) only received primary school education, whereas all younger smartphone users had at least received secondary school education. None of the smartphone users in this study were illiterate.

In terms of employment history, the majority of feature phone users had non-professional careers such as saloon worker, factory worker, tea lady, and tailor. In contrast, five out of the six smartphone users had professional careers such as nurse, clerk, teacher and secretary. It could be that these professionals were more financially stable, were living with pensioners' allowances and hence could afford to buy smartphones.

Further, we also investigated whether prior experience with different types of technological devices would influence smartphone adoption among the elderly. As expected, our study found that the elderly who used smartphones were more exposed to different kinds of technological devices, such as desktop computers, laptops and tablets. In contrast, the majority of non-smartphone users reportedly had none to very little experience with any similar technological devices.

B. Usage of Mobile Phone Functionalities

Table I summarizes phone functionalities used by the elderly. The table shows that apart from making calls, the most used function was texting messages. However, unlike younger mobile phone users who would be able to both send and reply text messages, not all of the elderly participants were able to do so. As shown in Table I, not all elderly participants who were able to read text messages were able to reply as well. When asked further on the reasons why they were not able to reply texts, some participants responded that it was due to their illiteracy, vision impairment, motor impairment, lack of knowledge, or they just felt 'lazy' due to the hassle of having to type the text messages. Most of the elderly preferred to directly make a call rather than to type a lengthy text.

[Participant P1, 62 years old, factory worker]:
"I don't like to read lengthy messages. It's hard to read."

[Participant P20, 65 years old, housewife]:
"I am only able to make calls with my mobile phone. I cannot reply to texts because I can't read. I did not go to school."

Table II indicates that most feature phone users used their phones only occasionally. Users of both multimedia phones

TABLE I. Usage of Mobile Phone Functionalities

Functionality	Feature Phone	Multimedia Phone	Smartphone
Call	12 (57%)	3 (14%)	6 (29%)
Read SMS	9 (43%)	2 (10%)	6 (29%)
Reply SMS	4 (19%)	1 (5%)	5 (24%)
Camera	6 (29%)	1 (5%)	6 (29%)
Photo Gallery	6 (29%)	2 (10%)	6 (29%)
Alarm	3 (14%)	1 (5%)	3 (14%)
Calculator	0	1 (5%)	1 (5%)
Games	3 (14%)	0	0
Calendar	1(5%)	1 (5%)	1 (5%)
Torch Light	2 (10%)	1 (5%)	0
Social Networking	0	0	0

TABLE II. Frequency of Mobile Phone Usage

Frequency of use	Feature Phone	Multimedia Phone	Smartphone
Very rare	1	0	0
Rare	0	0	1
Occasionally	8	0	0
Moderately	2	1	2
Often	1	2	3

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and smartphones, on the other hand, showed a higher frequency of usage with more than half stating that they used their mobile phones either moderately or often. Interestingly, when asked to elaborate further on what functionality they used most frequently, all of the elderly, regardless of the type of phone they used, stated that they used their mobile phone the most to call their children and family members. Furthermore, some even indicated that having a mobile phone is imperative especially when they were away from home, as it provided them with a sense of security that if an emergency happens, they can be quickly reached or contacted by their family members.

[Participant 1, 63 years old, shoe factory worker]:
"My life would be hard without my mobile phone. My husband is sick and stays home, so it is important to have my phone in case of an emergency especially when I am away from home. The mobile phone is indeed a necessity for me."

C. Barriers and Difficulties

To identify the barriers to smartphone penetration among older adults, we asked both feature phone users and multimedia phone users why they did not use more advanced smartphones. Table III summarizes the factors that prevent these elderly participants from using smartphones. Based on the table, the major factor that restricts them from using a smartphone is their perception that the smartphone is an expensive technological device that they cannot afford.

[Participant P8, 74 years old, housewife]:
"New smartphones are very expensive, yet are not even durable. It would be a big loss if it accidentally falls off my hand and crashes."

Besides that, age-related disabilities such as vision impairment also potentially inhibits the elderly from purchasing a smartphone, as they know it would be difficult for them as the font on the smartphone tends to be smaller than the feature phone's.

[Participant P20, aged 65, housewife]:
"My eyes cannot see clearly. I cannot read texts. Even to make a call, my children set a shortcut for me, like this." (Shows interviewer by pressing number 1 to call her first son, and number 2 to call her daughter).

Apart from that, some elderly were just not interested in having a more advanced type of mobile phone and some even perceived having a mobile phone as unnecessary.

[Participant 11, 72 years old, former caterer]:
"Actually my children wanted to give me a smartphone, but I said that I don't want it; I am too lazy to learn to use it".

Other factors like a lack of knowledge and mild cognitive impairments were somehow correlated. The elderly participants did admit that they lacked knowledge in using more advanced smartphones, but when asked further if they would like to use one if training was given, they seemed reluctant and unsure of themselves. They claimed that they were too old to learn, and even if they learned how to use it, they would forget the instructions over time due to mild dementia.

[Participant P19, aged 66, former factory worker]:
"I'm interested but I think I'm too old for that. My brain cannot accept it anymore. I would forget later."

As Table I indicates, smartphone users used the same functionalities as multimedia phone and feature phone users (i.e. only calls and SMSs). Hence it is important to understand what limits them to these basic functionalities. Table IV summarizes the difficulties that these elderly experience in using smartphones that inhibit them from utilizing the more advanced smartphone capabilities. Based on the table, elderly smartphone users have difficulties in using more advanced functionalities due to their limited knowledge or lack of training. Besides, some elderly have no interest and believe that it is not necessary to learn more advanced functionalities. As long as they are able to call their family members and friends they are satisfied.

Smartphone users among the elderly also have some issues with the smartphone interface due to mild vision impairments. One participant stated that he/she could hardly read words on his/her smartphone as the font was too small and hence he/she had to often look for his/her spectacles first before being able to read any SMS received.

[Participant P21, 64 years old, former clerk]:
"The problem with this smartphone is that the font is very small to read. My vision is blurry now, so it's quite difficult."

TABLE III. BARRIERS TO USING SMARTPHONE AMONG NON-SMARTPHONE USERS

Barriers	Feature Phone	Multimedia Phone
Not affordable	9	1
No interest	6	2
No necessity	6	0
Lack of knowledge	5	3
Vision impaired	5	2
Cognitive impaired	2	0
Hearing impaired	1	0
Motor impaired	0	1

TABLE IV. BARRIERS OF USING MORE ADVANCED FUNCTIONALITIES

Barriers	Number of participants
Lack of knowledge	4
Vision impaired	3
No necessity	3
Motor impaired	2
Not affordable	1
Lack of interest	1
Hearing impaired	0
Cognitive impaired	0

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Another design-related issue that the elderly have to deal with is the use of the touchscreen keypad. In our study, a few elderly participants suggested that due to their mild motor impairment such as difficulty in finger movement, using a touchscreen keypad to type an SMS was indeed tedious. These problems cause them to mistakenly send SMSs to a wrong person, or to send SMSs with a lot of typo-errors.

[Participant P3, aged 65, housewife]:
"I didn't know how to type SMS. The touchscreen keypad is really hard to use. The 'old phone' is much easier."

D. Lack of Interest and Unfamiliarity with Smartphone Usage

Apart from not being able to afford an expensive smartphone, another factor leading to low smartphone penetration among the elderly were lack of interest and the belief that having a smartphone is unnecessary. Further investigation revealed that the elderly in this study tended to use their mobile phones only for limited and basic functions, i.e. for calls and SMSs. Hence, they did not feel the need to purchase more expensive advanced smartphones when these two basic functions could already be done on a much cheaper feature phone.

[Participant P12, 72 years old, former coffer shop assistant]:
"The one that I'm using right now (feature phone) is good. As long as I can call my friends and family members it's good enough".

This, however, contradicts with [4], which found that the elderly desired a variety of applications beyond simply placing phone calls. In contrast, in this study, the elderly participants regarded ease of making a call as the most important and they were not interested in anything else. Earlier studies [5, 6], also show similar findings, stating that those aged 60 years and above only use their mobile phones for very limited purposes such as calling and texting in emergency situations.

Apart from only using mobile phones for basic purposes, another reason for the non-use of smartphones is that the elderly are already familiar with the older phones and have got used to the system. One participant clearly expressed her concern regarding this matter stating that if they wanted to use a smartphone, they would need to re-learn everything even for basic tasks like how to make a call or how to send an SMS. She stated that this would be hard as she perceived that she was already too old to learn new things and might forget what she had learned later.

These concerns are in line with [3] which found that lower adoption of mobile phones among elderly populations were due to difficulties in learning to use the system. Besides that, compared to the smartphone, the elderly in this study found that feature phones were more appealing to them due to their simplicity and rather straightforward functions [11].

These concerns were similar with [6] findings that the elderly tend to avoid using more complex functions. Ease of use and actual need of the services are indeed important criteria for elderly users. However, Culen and Bratteteig [11] study shows slightly contradictory findings as they state that the elderly's familiarity with a mobile phone is not an indicator that they like their phone. Rather, it was because they found the phone much easier and simpler to use.

E. The Effects of Prior Experience

Unlike younger adults, elderly smartphone users only used their smartphone for basic purposes, such as making calls and texting messages, similar to feature phone users. However, due to different interaction mechanisms between the smartphone and feature phone, they experienced different difficulties. One elderly participant who had been using the smartphone for about a year mentioned that she still experienced some difficulties in using the touchscreen keypad on her smartphone. She even credited the feature phone as being easier to use.

[Participant P21, 64 years old, former clerk]:
"It's hard to read on this advanced smartphone because the font is too small. Nokia (feature phone) is better and easy, easy to send SMS. Why? Because the keyboard is in the order of ABC to XYZ. On QWERTY keyboard, the alphabet is not in order, so it takes time for me to type an SMS. Plus, the font on Nokia is much bigger."

This clearly indicates that for the elderly, technological experience (experience in using feature phones) makes it harder to learn new devices (smartphones) because of the new interaction mechanism such as the use of a touchscreen, which needs to be re-learned. Reference [12] noted an interesting finding stating "Older adults have difficulty learning current UIs because they learned to use different types of UIs from older technologies in their formative years. As a result, the knowledge gained by older adults when using earlier UIs may not positively transfer to using existing computer UIs (touchscreen interface) and may make it harder for them to learn these UIs".

F. Dependency in Learning New Technology

Our study also found that some elderly participants were actually interested to learn more advanced smartphone functionalities but there was no one available to teach them. For example, one elderly woman who had a small personal business was interested to advertise her business on Facebook and access it through her mobile phone but she did not know how to do that, and no one was available to teach her.

[Participant 10, 60 years old, former clerk]:
"I'm interested to learn to use Facebook, Twitter and Internet on my smartphone, but nobody is available to teach me. I think it would be beneficial for me to know these things, so that it is easier to connect with my customers."

Further as shown in Table IV above, some elderly who refused to use smartphone due to 'lack of knowledge' stated that they were interested to learn if there were enough resources available i.e. financial resources and training.

[Participant P4, 73 years old, former tea lady]:
"Actually I am interested to learn to use a smartphone, but it's expensive and I cannot afford it. But if there is enough money, I might want to try it."

[Participant P5, 62 years old, former printing factory worker]:

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"If there is someone who wants to teach me, then I am interested to learn."

The above quotations indicate that some elderly do have an interest to learn but they need somebody to teach them. This shows that in learning new devices, the elderly tend to depend on other people. Reference [5] found similar finding where the elderly generally learn to use mobile devices with their friends or children, and not alone. Reference [3] also stated that older adults in their study expressed wanting to take a class to learn to use their mobile devices.

V. LIMITATIONS

There were very limited numbers of participants in this study. There is therefore a need to conduct studies with larger samples of the elderly. With more information on how the elderly use their mobile phones, we will be able to design mobile systems that will be able meet their needs [4].

Besides, as the interviewer spoke a different language from some of the elderly participants, there was language barrier experienced during data collection. In this study, interviews were done in the Malay language. Although the language is regarded as the national language, only a few of the Chinese elderly participants were able to communicate fluently and understand the questions well. Others experienced some difficulty in understanding the questions being asked, and as the interviewer had to revise the questions, the intended questions may have been slightly different. Furthermore, as some of them had difficulties in communicating in Malay, they spoke very little, and hence very little insights could be gathered. Thus, it would be very beneficial if the interviewer can speak the same language as the elderly, so that more useful and valid data can be gathered.

VI. RECOMMENDATIONS

Although there were diverse studies investigating smartphone usage among the elderly, we found that most of these studies only focused on application-specific usability issues. We realized that not many studies actually try to understand in-depth why many elderly refused to shift from traditional feature phone to more sophisticated smartphone in the first place. We argue that this is very imperative issue for designers to understand particularly at the early phase of application development where user needs and requirements were accessed. We believe conducting in-depth research to understand the barriers of smartphone usage among the elderly would productively assist designers in developing application that would able to cater the needs of this vulnerable population.

VII. CONCLUSIONS

Among the major barriers that prevent the elderly from using smartphones are financial restrictions, vision impairment and lack of knowledge in using smartphone functionalities. Besides, due to lack of information on the diverse beneficial aspects of smartphone usage, the elderly tend to perceive both feature phones and smartphone functions as similar and hence they were not willing to face the difficulty of typing on small touchscreen keyboards, or spending a lot more money on the smartphone for something that they can do on cheaper, more easily-operated devices like the feature phone. Thus, we believe that the elderly would be most likely to adopt smartphones if they found the smartphone not only easy to use but also beneficial to them.

ACKNOWLEDGEMENT

We would like to thank the elderly participants involved in this study. The work was supported by the Ministry of Science, Technology and Innovation (MOSTI) research grant (06-01-02-SF1083).

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